

✓ Task 1

```
import os
import cv2
import matplotlib.pyplot as plt
```

```
import kagglehub

# Download latest version
path = kagglehub.dataset_download("paultimothymooney/blood-cells")

print("Path to dataset files:", path)
```

Using Colab cache for faster access to the 'blood-cells' dataset.
Path to dataset files: /kaggle/input/blood-cells

```
!cp -rf /root/.cache/kagglehub/datasets/paultimothymooney/blood-cells/versions/6
```

```
image_path = "/content/dataset-master/dataset-master/JPEGImages/BloodImage_000000.."
```

```
image = cv2.imread(image_path)

# Convert BGR to RGB
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
gray = cv2.cvtColor(image, cv2.COLOR_RGB2GRAY)
```

```
_, prediction = cv2.threshold(
    gray,
    0,
    255,
    cv2.THRESH_BINARY + cv2.THRESH_OTSU
)
```

```
plt.figure(figsize=(10,5))

plt.subplot(1,2,1)
plt.imshow(image_rgb)
plt.title("Original Blood Cell Image")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(prediction, cmap="gray")
plt.title("Simulated Prediction (Binary Mask)")
plt.axis("off")
```

```
plt.axis('off')
```

```
plt.show()
```

Original Blood Cell Image



Simulated Prediction (Binary Mask)



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